```
clc;
clear;
close all;
Am=1;
Ac=1;
fm=5;
fc=50;
beta=5;
kp=5;
fs=1000;
t=0:1/fs:1;
m = Am*cos(2*pi*fm*t); % Message signal

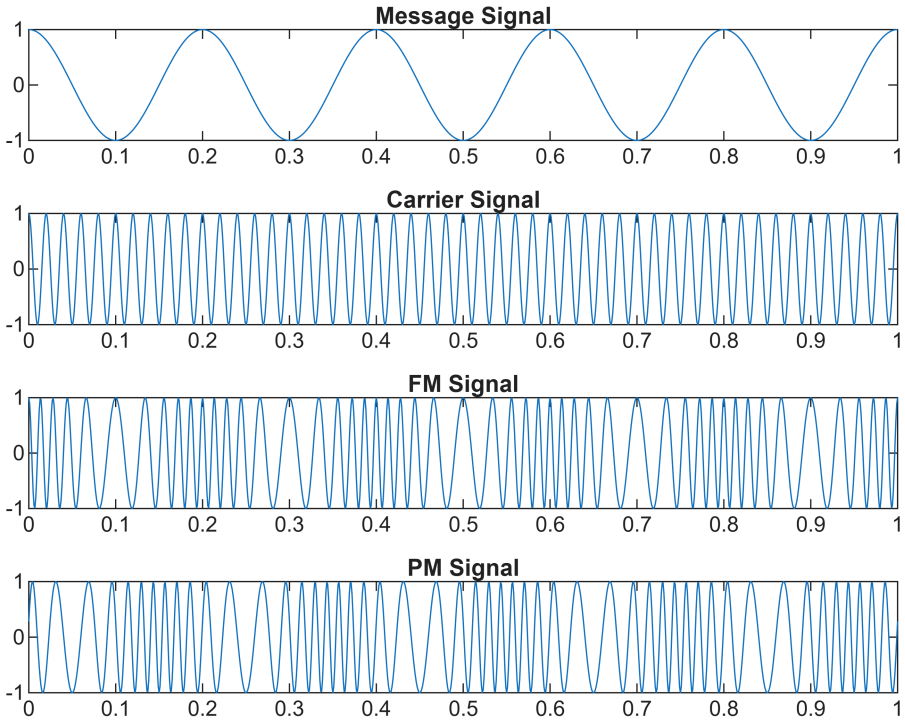
c = Ac*cos(2*pi*fc*t); % Carrier signal
fm_sig = Ac*cos(2*pi*fc*t + beta*sin(2*pi*fm*t)); % FM
pm_sig = Ac*cos(2*pi*fc*t + kp*m); % PM
Df = beta*fm; % Frequency deviation
fprintf('Frequency Deviation = %d Hz\n',Df);
```

```
Frequency Deviation = 25 Hz
```

```
fprintf('FM Modulation Index = %d\n',beta);
```

```
FM Modulation Index = 5
```

```
subplot(4,1,1), plot(t,m), title('Message Signal')
subplot(4,1,2), plot(t,c), title('Carrier Signal')
subplot(4,1,3), plot(t,fm_sig), title('FM Signal')
subplot(4,1,4), plot(t,pm_sig), title('PM Signal')
```


Message Signal
Carrier Signal
FM Signal
PM Signal
1
0
-1
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1